

Agentic AI in the Life Sciences

From Autonomous Discovery to Private, Local Deployment

Block 4 of 4 – Applied Perspectives – KDD 2026 Tutorial, Jeju Island

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Based on joint work with Grace Hui Yang (Georgetown), Pranav N. Venkit (Salesforce), Hooman Sedghamiz (Bayer), Enrico Santus (Bloomberg), Victor Dibia (Microsoft Research), and Ioana Baldini (Bloomberg) – presented today by Enrico Santus, Grace Hui Yang, and Hooman Sedghamiz.

KDD 2026

Closing block of “Agents in the Wild: Where Research Meets Deployment” (KDD 2026 Tutorial).

Hooman Sedghamiz

Sr. Director, AI/ML – Insights, **Bayer AG**

- Leads enterprise-scale **generative AI platforms for precision medicine** serving 100,000+ employees globally.
- Research at the intersection of **medical NLP, biomedical AI, and pharmacovigilance**; 550+ citations across EMNLP, BMC, Frontiers.
- Co-Chair, **EMNLP 2023 GEM Industrial Track**. Work spans medical NLP, computational biology, and biomedical signal processing.
- Focus today: bridging **academic agentic research** with **real pharmaceutical deployment**.



Hooman Sedghamiz

Where this session sits

The tutorial arc – four blocks, three speakers

Block 1: Foundations — reasoning, planning, coordination. **Block 2:** Retrieval Pipelines & Evaluation Beyond Benchmarks. **Block 3:** Agents in Finance. **Block 4 (this session):** Agents in Pharma & Life Sciences, and Deployment.

Why life sciences is the testbed

- high-value, high-stakes decisions
- rich literature + structured data + wet labs
- a hard, non-negotiable **truth boundary**: the experiment
- strict **privacy** constraints on patient data

Two movements today

- **Discovery:** autonomous multi-agent science (Robin & the landscape)
- **Deployment:** a production deep agent, trajectory-level QA, and the constraints that decide what ships

The through-line

Not how autonomous can it be but under what constraints can it act reliably — and where must the data, and the human, stay?

Learning objectives

By the end of this session, you should be able to:

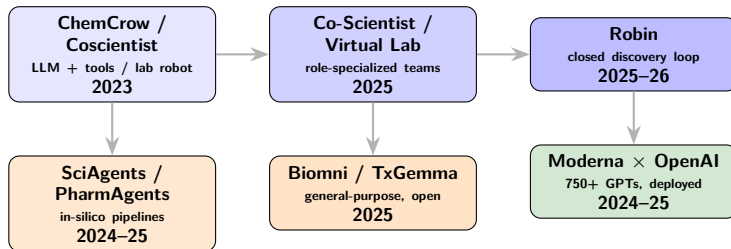
- 1 **Map** the landscape of agentic AI for biomedical discovery.
- 2 **Trace** an end-to-end multi-agent discovery loop (FutureHouse Robin) and locate the human-in-the-loop boundary.
- 3 **Distinguish** validated, narrow wins from broad in-silico claims — and why wet-lab + expert review remain the ground truth.
- 4 **Apply** RL with verifiable rewards (GRPO) to train a small domain reasoner, and say when a checkable reward exists.
- 5 **Explain** how a research-grade architecture becomes a production system, and why the trajectory is the unit of QA.
- 6 **Name** the open research challenges and industrial constraints — latency, compute, monitoring — that decide what can ship safely.

Sources: [Yang et al., 2026].

Roadmap

- 1 Part 1: The Pharma & Life Sciences Discovery Landscape
- 2 Part 2: Robin – An End-to-End Discovery Loop
- 3 Part 3: The Agent's Scientific Toolbelt
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From single agents to virtual labs (2023 → 2026)



Reading the shift

One LLM + tools (or a lab robot) → **teams** of role-specialized agents that debate, critique, and rank
→ closed loops with wet-lab validation, and in parallel **enterprise deployment** at industrial scale.

Sources: ChemCrow [Bran et al., 2023]; Coscientist [Boiko et al., 2023]; AI Co-Scientist [Gottweis et al., 2026]; Virtual Lab [Swanson et al., 2024]; Robin [Ghareeb et al., 2026]; Moderna [OpenAI, 2025].

A taxonomy of life-science agents

Bucket	What they do	Examples
(a) Hypothesis generation	propose & rank hypotheses	Co-Scientist, SciAgents, Virtual Lab, Robin
(b) Lab automation / embodied	<i>physically run</i> experiments	Coscientist (Opentrons)
(c) Molecular / therapeutic design	design molecules / biologics	ChemCrow, PharmAgents, TxGemma, Virtual Lab
(d) Literature / general-purpose	reason over the biomedical corpus	Biomni, Owkin K Navigator, PaperQA
(e) Enterprise platforms	agentic AI in production	Moderna × OpenAI (750+ GPTs)

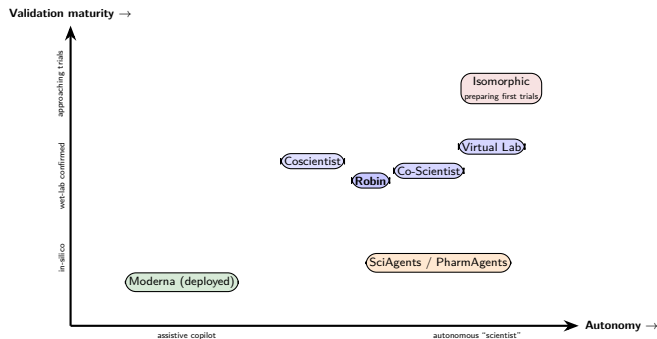
The shared blueprint

Planner/PI + Critic + Ranking + specialists + **deterministic scientific tools** (AlphaFold-Multimer, Rosetta, ESM, RDKit, retrosynthesis, lab robots). *The LLM orchestrates; the tools do the science.*

Sources:

[Gottweis et al., 2026, Ghafarollahi and Buehler, 2025, Swanson et al., 2024, Boiko et al., 2023, Bran et al., 2023, Gao et al., 2025, Google DeepMind, 2025, Huang et al., 2025, OpenAI, 2025].

Two axes that organize the field



The honest reading

- Credible wins (upper-middle/right): **narrow + wet-lab-validated**.
- Boldest autonomy claims often sit **in-silico only**.
- Scale deployment (Moderna) is real but **assistive**, not autonomous discovery.

Sources: Positions synthesized from the cited primary sources. [Ghareeb et al., 2026, Swanson et al., 2024, Gottweis et al., 2026, OpenAI, 2025, Isomorphic Labs, 2025].

Validated wins are real — but narrow

Worked in a wet lab

- **Coscientist:** Suzuki + Sonogashira on a real robot — GC-MS confirmed.
- **Virtual Lab:** 92 SARS-CoV-2 nanobodies validated; mutants *reported* to bind JN.1 / KP.3; ~1% human-authored (*per paper*).
- **Co-Scientist:** AML repurposing confirmed in cell lines; fibrosis targets active in organoids (*collaborator report*).
- **Robin:** surfaced **ripasudil** for dry AMD (Part 2).

The caveat

- **ChemCrow:** GPT-4-as-judge *couldn't tell* wrong answers from correct tool-grounded ones.
- $\Rightarrow$ LLM self-eval / Elo tournaments are **no** substitute for expert + wet-lab truth.
- “surpasses human research” / “20 $\times$ faster” claims are mostly **in-silico**.

Where humans stay in the loop

Across every validated system, human-and-wet-lab sits at the **truth boundary**: agents generate, rank, refine; humans set the goal and run the deciding experiment.

Sources: [Boiko et al., 2023, Swanson et al., 2024, Gottweis et al., 2026, Bran et al., 2023].

Enterprise reality check: Moderna × OpenAI

Agentic AI *deployed*, not just demoed

- **750+ custom GPTs** in ~2 months — legal, research, manufacturing, commercial.
- **“Dose ID”** GPT evaluates optimal vaccine dose: reasoning, sources, charts.
- ~120 conversations / user / week; toward ~5,000 licenses.

Why it matters here

- The “boring but real” counterpoint to research demos.
- **Human-operated copilots** — employee-in-the-loop, not autonomous discovery.
- Value is **operational**: adoption, throughput, governance — not one headline molecule.

Two ends of the same field

Research prototypes chase **autonomous discovery**; enterprises ship **governed assistance**. The honest story needs both columns.

Sources: [OpenAI, 2025].

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Robin (FutureHouse): AI mind, human hands

What Robin is

A **multi-agent system** that automates the *intellectual* loop of experimental biology — hypothesize → design experiment → analyze data → revise — by orchestrating three specialized agents. Humans run every bench experiment.

Three agents, one loop

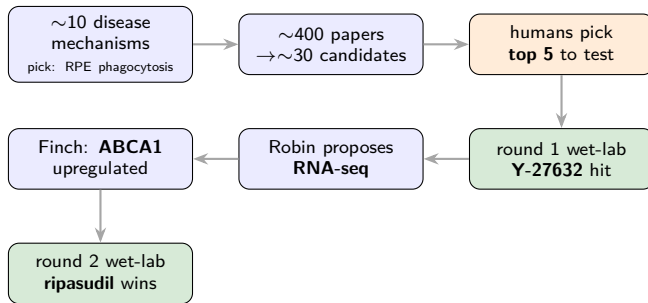
- **Crow** — proposes hypotheses and strategy from the literature
- **Falcon** — reviews and ranks therapeutic candidates
- **Finch** — writes and runs code to analyze wet-lab data

Candidate ranking = an **LLM-judged tournament** (rationale, pharmacology, literature support).

Deliberately minimal

- “an interpretable workflow of only three agents”
- literature agents built on the **PaperQA2 / Aviary** stack
- seed with *only a disease name*; open sourced (Apache-2.0)
- ecosystem context: **ether0**, a 24B chemistry reasoning model

The discovery funnel: dry AMD → ripasudil



The result

Ripasudil — a ROCK inhibitor approved for glaucoma — **had never (to the authors' knowledge) been proposed for dry AMD**. Robin linked ROCK-inhibition biochemistry to retinal (RPE) biology — a cross-disciplinary jump a specialist would be unlikely to make. *All hypotheses, experiment choices, analyses & main-text figures: Robin-generated.*

Bounded autonomy, literally

Robin did autonomously (the “mind”)

- generated all hypotheses & the mechanism strategy
- designed / proposed the experiments (incl. the follow-up RNA-seq)
- analyzed all data (Finch wrote & ran its own code)
- produced **all main-text figures**

Humans did (the “hands” + the gate)

- ran **every** physical wet-lab experiment
- **selected which candidates** to test (top 5 of ~ 30)
- verified the analyses, formatted figures

Where the AI analysis was wrong

Key result: Finch (AI) reported **7.5-fold**; human re-analysis found **1.75-fold** — same direction, $\sim 4\times$ gap. **AI analysis $\neq$ ground truth.** Preclinical only; a narrow *repurposing* problem.

Concept-to-paper-*submission*: **2.5 months**. Blog says “entirely AI-driven”; paper says “**semi-autonomous**” — both stress a scientist stays in the loop.

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Tools agents can call *now*: Boltz

An open, agent-callable structure + affinity tool

- **Boltz-1**: first *fully open source* model to **approach AlphaFold3 accuracy** for biomolecular complexes.
- **Boltz-2**: jointly predicts **structure + binding affinity** — “the first DL model to **approach FEP accuracy** while running **~1000× faster**.”
- pip-installable **CLI**; input is a **YAML** (or FASTA) → structure + affinity outputs.

Open vs. closed

- **✓ Boltz** — **MIT license** (academic *and* commercial), weights + CLI in the open.
- **✗ AlphaFold3 / Isomorphic** — closed weights; not droppable into your own loop.

Why MIT + CLI + structured I/O matters

A permissive license, a one-line invocation, and machine-readable outputs make Boltz a **deterministic, reproducible step** an agent can call inside a drug-discovery loop.

Sources: Boltz README, <https://github.com/jwohlwend/boltz> (Boltz-1/Boltz-2 claims, MIT, CLI/YAML I/O); [Isomorphic Labs, 2025] (closed-weights contrast).

Boltz as a tool call: predict, then read affinity

The invocation the agent runs

```
# YAML in -> structure + affinity out
boltz predict input.yaml \
    --use_msa_server
```

What comes back (per ligand)

```
affinity_probability_binary # 0-1
affinity_pred_value        # log10(IC50)
```

How the agent uses it

- Propose a ligand → call boltz predict → **read back** the affinity outputs → iterate.
- affinity_probability_binary **triages binders vs. decoys** (hit discovery).
- affinity_pred_value ($\log_{10}(\text{IC}_{50})$) drives **lead optimization**.

The model is a tool, not the brain

The model becomes a **deterministic tool**, not the agent's brain: easy to log, cache, and unit-test.

Sources: Boltz README, <https://github.com/jwohlwend/boltz> (boltz predict -use_msa_server; affinity_probability_binary = binder/decoy, affinity_pred_value = $\log_{10}(\text{IC}_{50})$).

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One method, one principle, three scales

Reasoning is *trained*, not prompted

It **emerges** from reinforcement learning on answers a **program can check** — RL with verifiable rewards (RLVR). No human preference labels, no reward model to game.

Frontier

- DeepSeek-R1
- GRPO + rule-based accuracy & format rewards on math / code / STEM.
- Reasoning emerges from RL alone (R1-Zero, **no SFT**).

Domain specialist

- **ether0** — chemistry, 24B.
- Same recipe, verifiable chemistry rewards.
- ✓ **640,730** problems / **375** tasks; beats frontier models *and* human experts on molecular design.

Do-it-yourself

- **Medical mini-R1** — medicine, 3B.
- Same pattern, miniaturized: GRPO on a **\$3 Colab T4**.
- Verifiable medical answers (PubMedQA / MCQ / GSM8K).

Why RLVR fits medicine

Many medical answers are **checkable** — MCQ, PubMedQA yes/no/maybe, dosage bounds $\Rightarrow$ a **cheap, objective, hack-resistant** reward. The same “verifiable truth” anchor as the wet-lab loop in Part 2.

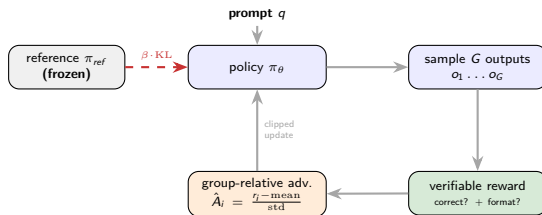
GRPO in one slide

Group Relative Policy Optimization

- **No critic / value network** — the **group** of sampled outputs *is* the baseline.
- Per prompt: sample **G** outputs, score each with a reward r_i .
- Advantage: $\hat{A}_i = (r_i - \text{mean}(r)) / \text{std}(r)$ over the group.
- Penalize drift with $\beta \cdot \text{KL}$ to a **frozen reference**; clipped policy update.

	PPO	GRPO
Baseline	learned critic	group mean
Extra model	✗ yes	✓ no

Sources: GRPO [Shao et al., 2024] (Eq. 3); [Suma and Dauncey, 2025]; TRL GRPOTrainer.



No value network — the group is the baseline.

Build your own medical mini-R1

The recipe

- Base **Qwen2.5-3B-Instruct**.
- **QLoRA + Unsloth + vLLM**; TRL GRPOTrainer.
- Data: **PubMedQA + GSM8K + specialty MCQs**.
- Output forced into
`<reasoning>..</reasoning>`
`<answer>..</answer>.`
- **Verifiable rewards**: correctness 2.0 / 0 + XML format 0.5.

Caveat

A reproducible *tutorial*; results are qualitative — **not** a benchmarked clinical model.

```
import re
from trl import GRPOConfig, GRPOTrainer

def pull(t):
    m = re.search(r"<answer>(.*?)</answer>",
                  t, re.S)
    return m.group(1).strip().lower() if m else ""

# verifiable correctness reward
def correct_reward(completions, label, **kw):
    outs = [c[0]["content"] for c in completions]
    return [2.0 if pull(o) == gt else 0.0
            for o, gt in zip(outs, label)]

# format reward (R1-style structure)
PAT = r"<reasoning>.*?</reasoning>\s*" \
      r"<answer>.*?</answer>"
def format_reward(completions, **kw):
    outs = [c[0]["content"] for c in completions]
    return [0.5 if re.search(PAT, o, re.S)
            else 0.0 for o in outs]

cfg = GRPOConfig(
    learning_rate=5e-6,
    num_generations=6,      # group size G
    max_steps=750, use_vllm=True)
trainer = GRPOTrainer(
    model="Qwen/Qwen2.5-3B-Instruct",
    reward_funcs=[correct_reward,
                  format_reward],
    args=cfg, train_dataset=ds)
trainer.train()
```

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The privacy problem the demos skip

Discovery agents read papers. Clinical agents read patients.

The moment an agent touches **Protected Health Information** — clinical notes, path reports, discharge summaries — a cloud API call can become a **third-party disclosure**.

Send raw notes to a cloud LLM

- HIPAA Business Associate Agreement
- GDPR special-category health data + cross-border transfer
- EU AI Act obligations
- opaque vendor, shifting terms, per-call cost

Run the model locally / on-prem

- **no data transmitted** $\Rightarrow$ no BAA, no transfer question
- air-gap friendly; auditable, pinned, reproducible
- open weights $\Rightarrow$ no per-call fee, no telemetry
- de-identify *at the source*, then (optionally) send only scrubbed text onward

The most private API call is the one you never make.

Before / after de-identification (runs locally)

A local, on-device de-id model (149M, ModernBERT, **micro-F1 ≈ 0.95**)

Before

Patient John Smith (DOB: 03/15/1985,
SSN: 123-45-6789), john.smith@email.com,
(555) 123-4567.
456 Oak Street, Boston, MA 02108.

After

Patient [FIRST_NAME] [LAST_NAME]
(DOB: [DATE_OF_BIRTH], SSN: [SSN]),
[EMAIL], [PHONE_NUMBER].
[STREET_ADDRESS], [CITY],
[STATE] [POSTCODE].

Why de-id belongs at the first hop

- De-id is a **deterministic local tool**, not the agent's brain — easy to log and unit-test.
- **De-id as the first hop** = the **privacy gate**; only scrubbed text ever flows onward to any reasoning step.
- Covers a *superset* of HIPAA's 18 Safe Harbor identifier categories.

Honest limitation

Automated de-id **reduces, not eliminates** re-identification risk. **Keep a human in the loop**; validate on your own data before trusting it as a boundary.

Sources: OpenMed de-id model family [Panahi, 2025]; the micro-F1 is **project-reported** on a synthetic PII test split, from the model card — not an independent clinical benchmark. HIPAA Safe Harbor.

OpenMed: the open, local-first medical model stack

"Your Data. Your Model. Your Hardware." — the de-identification model on the previous slide is OpenMed's.

What it is

- **Apache-2.0** weights that run **100% on-device** — no cloud call, no patient data leaving the network.
- **1,000+** medical models on Hugging Face (the org lists ~1,500), with hundreds of millions of downloads.
- **Clinical NER** across dozens of entity types: disease/condition, medication, anatomy, gene & protein.
- **HIPAA PII de-identification** ("Privacy Filter") incl. a multilingual variant.
- On Apple Silicon, the MLX build of the Privacy Filter family is **20–30× faster** than CPU PyTorch.

Why it matters here

- Open weights + permissive license + on-device execution turns **privacy from a policy promise into an architecture**.
- For extraction and de-identification, a **small specialist beats a frontier generalist**: pinnable, cheap, unit-testable, auditable.
- It is why the de-id gate in the demo runs **with the network off**.

Credit

An **independent open-source project**, used here as an inspiration and a dependency. **Not a Bayer product**.

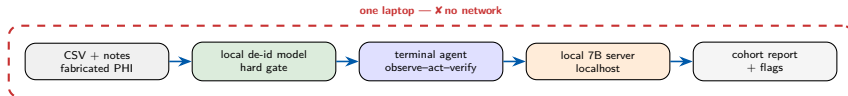
Sources: OpenMed [Panahi, 2025]; <https://github.com/maziyarpanahi/openmed>. Counts and speedups are project-reported.

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A local agent on a laptop — what the run produced

A terminal agent driven by a **local 7B model** was pointed at a folder of **fabricated** medical records. It de-identified them, computed cohort statistics, flagged data-quality problems and wrote a report — **with the network off**. Below is its actual output.



De-identification, on-device

John Carter (MRN 558231), seen
2025-03-04 by Dr. Alvarez ...
Phone (555) 274-9981.

[NAME] (MRN [MRN]), seen
[DATE] by Dr. [PROVIDER] at
[FACILITY]. Phone [PHONE].

Cohort summary + auto-flagged issues

rows: 23 T2D: 11 | HTN: 8 | HLD: 4
mean A1c: T2D 8.27 | HTN 5.90
A1c >= 8.0: 7 patients
flags:
[MISSING] P003 no systolic_bp
[RANGE] P021 age = 210
[DATEFMT] P022 03/17/2025

The number that matters: 0 bytes on the wire

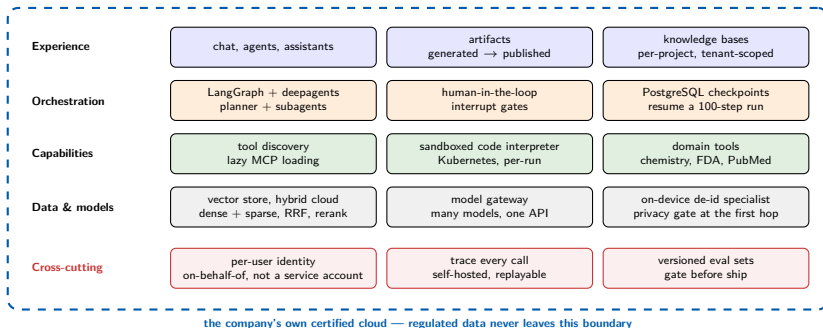
The network monitor stayed at **zero** for the whole run. PHI loading, de-identification, analysis and report writing all happened on the machine — so the privacy claim is a property of **where the compute ran**, not of a policy document.

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What it takes to put an agent into production

The research prototype is **one box** in this picture. The rest is what a regulated deployment adds — and where the engineering actually goes.

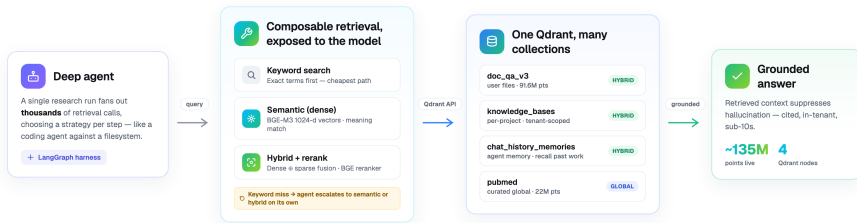


The lesson from two years of running it

Self-hosting met compliance but the ops load crushed a small team — a **hybrid** control plane fixed that. The red row is no add-on: **retrofitting identity, tracing or evaluation after launch is the expensive path.**

Retrieval the agent drives itself

One deep-research run unrolls a tool-calling loop **hundreds of steps deep**.



The vector-search API is exposed directly to the model

- human or agent, **same interface**: keyword → semantic → hybrid
- knowledge bases mount as a virtual filesystem /kb/{id}/: ls, read_file, grep, glob
- grep semantic:... dispatches to hybrid search, chosen **at runtime**

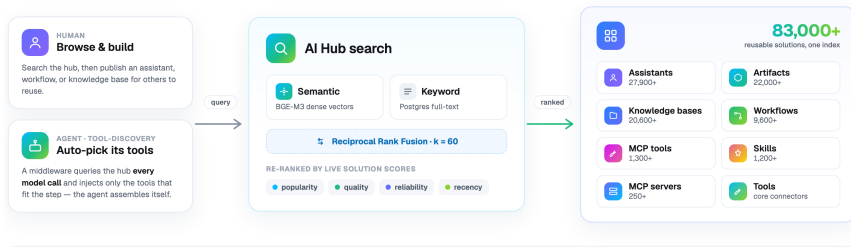
Parallel query expansion

- one question fans into **four** simultaneous searches: verbatim, question-form rewrite, keywords, HyDE hypothetical answer
- underneath: dense + sparse in parallel, fused with **RRF**, sharpened by a **BGE reranker**
- why four: the same concept appears under **different nomenclature** in regulatory filings, clinical protocols and marketing copy

Sources: Bayer myGenAssist platform. Figures and diagram: first-party.

One search fabric for people and agents

The assets worth finding stopped being files: they became what people **built**. Build one, and **every colleague and every agent** can find it.



One index, two kinds of caller

- every solution carries a **1024-dim BGE-M3** embedding, fused with full-text matching via **RRF**
- humans lean on it: **~49,000 searches/month** across **7,500+** distinct people

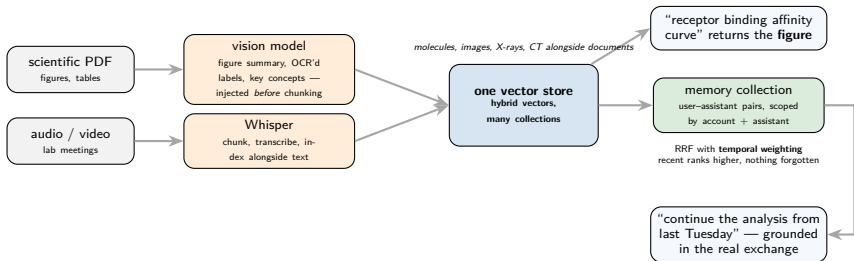
Agents use the same search to equip themselves

- a **discovery middleware** injects only the **~12 tools** that fit the step, out of a **100+** tool surface — **~70%** fewer prompt tokens, nothing unreachable
- the agent can find a specialist assistant and delegate to it as a **subagent**

Sources: Bayer myGenAssist AI Hub. Figures and diagram: first-party.

Every modality, remembered — and read only once

Life-sciences evidence is not text, and a deep run fires **thousands** of retrievals. Both facts shape the store — which is quietly becoming **the company's search engine**.



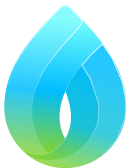
Omnimodal — figure descriptions enter the text stream *before* chunking, so the figure itself is retrievable; molecules, X-rays and CT sit beside documents, and a lab meeting is searchable **minutes after upload**.

Memory — when a user returns weeks later the agent **proactively searches** prior exchanges, ranked by RRF with **temporal weighting**; a nightly job backfills gaps. Continuity, not a summary.

Read once — scraped chunks are written **back into a collection** instead of re-paying the external API, and heavy OCR runs **on demand**, then is cached for everyone: **~80% less compute**.

Open problem: a cached answer goes **stale** the moment the source changes — deciding when to re-fetch, without re-fetching everything, is unsolved.

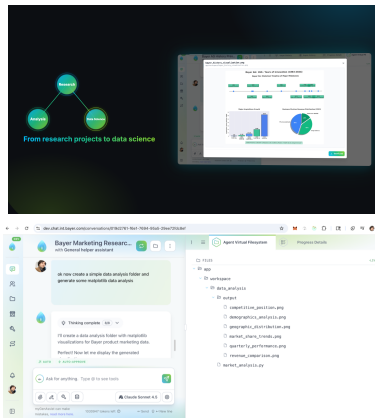
A deep agent, in production



Planner, subagents, approval gates — shipped to 116,000+ employees (Bayer-internal).

- **Planner + subagents** — a researcher and an artifact-creator
- **Human approval** pauses risky actions before they run
- **Resumable state**: every step is saved and replayable

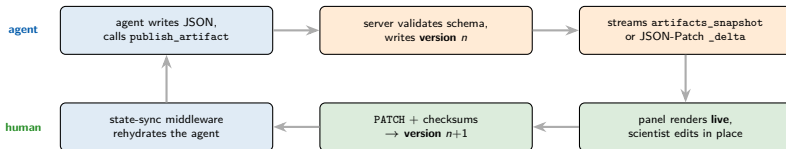
Sources: Bayer myGenAssist platform (architecture). User count is a Bayer-internal figure.



The agent plans, writes code, and runs it in an isolated sandbox.

Human and agent editing the same object

Approval gates stop the agent *before* a risky action. The **artifact** is how a scientist corrects what it has *already* produced — a typed, versioned object, edited in place.



An object, not a message

- 14 typed kinds — charts, tables, documents, slides, code — schema-validated on publish
- deltas ship as **RFC-6902 JSON Patch**, so the panel repaints **while the agent is still writing**

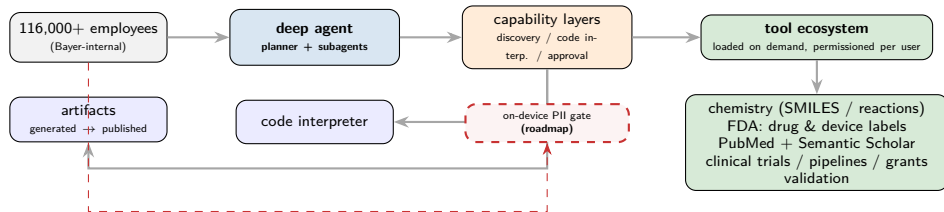
The human is a first-class editor

- React artifacts **run in a sandbox**: review is of the working result, not the code
- **checksums** catch the agent moving underneath, and the **human's edit wins**

Why this matters in a regulated field

Every write from either side is an **immutable version tagged with its origin** — *who changed this figure before it reached the report* has an answer. And nobody pastes the correction back into chat: the next turn loads the latest state into the agent's workspace, so **the expert's correction is the context** the agent continues from.

A life-science tool ecosystem



One agent, many domains

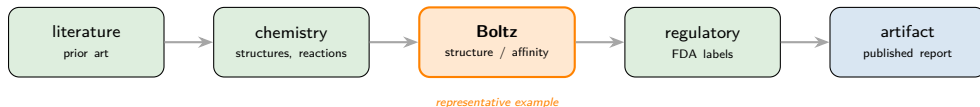
Tools **load on demand** and are **permissioned per user**; chemistry, FDA, PubMed, and clinical-trials tools sit around a shared code interpreter and artifact pipeline. The dashed box is **on the roadmap** — today, privacy protection is server-side (pseudonymization on deletion; automated scrubbing in monitoring).

Sources: Bayer myGenAssist platform (architecture). On-device PII gate is on the roadmap.

Extending the agent: drop in a new tool

Add a capability, change nothing in the agent core

A new tool registers once and **inherits the whole stack**: tool discovery, the code interpreter, human-approval gating, per-user permissioning, and tracing.



A drug-discovery trajectory

One run chains domains end-to-end: **literature** → **chemistry** → a structure-prediction tool like **Boltz** → **regulatory** context → a published **artifact**. Read-only tools can be chained automatically by the code interpreter; tools that change state require human approval first.

Sources: Bayer myGenAssist platform (architecture). Boltz shown as a representative drop-in example, not currently shipped.

Agent QA: the trajectory is the unit of analysis

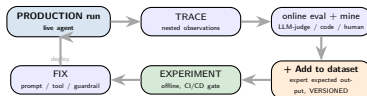
Agents are **non-deterministic and multi-step**: the same input yields different tool calls each run. A single final-answer score is **blind to *how* the agent got there**.

Every run = a replayable trace (Langfuse [Langfuse, 2025])

- **Session** → **Trace** → nested **Observations**
- typed steps: generation / tool / retriever / agent / guardrail
- each step logs inputs, outputs, timing, cost
- the nesting tree *is* the trajectory

Why it fits medicine

- **open source (MIT)**, OpenTelemetry-based
- **self-hostable on-prem** / VPC / air-gappable
- ⇒ **PHI never leaves the boundary**
- auditable + reproducible by construction



Production failures become permanent regression tests

Failing trajectories become a **versioned** eval dataset; experiments gate the next deploy — so the same failure **can never** silently recur.

Where this is going: a trace for every agent call

Every agent call needs a trace — and in a regulated industry the **user**, not just the platform team, gets to open it.

What already ships

- **Inline clickable citation** per retrieval hit: exact chunk text, source document, deep link into the knowledge-base viewer — **editable in place**.
- **One structured trace per session**, self-hosted on-prem: cost per model call, latency per hop.
- **Operational metrics**: latency percentiles, circuit-breaker transitions, indexing throughput.

Direction of travel

- Platform-team observability → **user-facing transparency**: the user sees the **fleet of agents** that worked for them, and opens any one.
- **Traces as first-class data**, shipped next to the output — not buried in a backend log.
- **GxP-grade reproducibility as a product surface**.

The auditor test

A regulator asks *how was this answer generated?* — and the platform reconstructs **which collections were queried, which chunks scored highest, which model produced the synthesis**. If you cannot rebuild that path, the answer is not evidence.

2026 ambition: agentic **drug-discovery** pipelines; **chemical research** workflows with scientists in the loop; **regulatory preparation** — agents assemble submission packages, every claim grounded and every citation verified.

Sources: Bayer myGenAssist roadmap (first-party). Wider industry direction — decisions traceable to intent, outcomes as first-class data, organizations keeping sovereignty [Entire, 2025].

Why it matters in medicine

- **High stakes** $\Rightarrow$ an auditable, step-by-step record of every decision.
- **Reproducible** $\Rightarrow$ versioned datasets and prompt versions = re-derivable evidence.
- **PHI stays in the room** $\Rightarrow$ traces self-hosted on-prem; the patient's data never leaves the boundary.
- **Attributable** $\Rightarrow$ regulated content lives in versioned, access-scoped collections — clinical guidelines, adverse events, trials, regulatory affairs — so every retrieval points back to one of them.
- **Answerable to an auditor** $\Rightarrow$ which collection, which chunk, which model can be reconstructed after the fact.

The through-line

“If you can't replay it, you can't debug it” — and in medicine, those traces stay on-prem. **Latency, compute efficiency, and monitoring** are not afterthoughts; they are load-bearing parts of a safe deployment.

Sources: Langfuse (open source, self-hosted) for on-prem tracing. Bayer myGenAssist platform.

Where this is going: train the agent, don't just prompt it

Today we **prompt** a general model and wrap it in scaffolding. Next: **train** it — and let the platform do the training.

What Part 4 established

- The GRPO recipe works at **every scale** — frontier, a 24B chemistry specialist, a **3B medical reasoner on a \$3 T4**.
- The only hard requirement is a reward a **program can check**. Where that exists, reasoning is trained rather than prompted.
- The question is no longer *can we train a domain reasoner*, but **which loop generates the signal**.

The ambition: reward the trajectory

- Train on the **whole research run** — which tool, which query, which experiment next — not the final token alone.
- Drug development supplies its own verifiable rewards: docking scores, assay readouts, retrosynthesis feasibility.
- **AutoResearch**: the agent runs the loop — proposes the task, mines its traces, trains a specialist, keeps it only if it wins held-out.



Traces, versioned eval sets and a searchable model hub are what that loop needs — **myGenAssist ships all three today**.

Sources: [Sedghamiz, 2025, Suma and Dauncey, 2025, Narayanan et al., 2025]. mini-R1 is a reproducible tutorial with qualitative results, not a benchmarked clinical model.

Open research challenges and industrial constraints

The same open problems cut across every domain agents are pushed into. None is fixed by a stronger base model alone.

Open research challenges

- **Multi-agent scalability** — coordination cost as the number of agents grows
- **Lifelong adaptation** — agents that keep learning safely after deployment
- **Explainability of collective reasoning** — why the *team* reached a conclusion
- **Secure inter-agent communication** — protecting the channels agents use to coordinate
- **Embodied or multimodal agents** — reasoning beyond text, into the physical world

Industrial constraints

- **Latency** — multi-agent loops are slow; clinicians will not wait indefinitely
- **Compute efficiency** — every extra agent, retry, or tool call has a real cost
- **Monitoring** — production agents need trajectory-level tracing, not endpoint metrics

What this asks of research

Academic rigor in safe, reliable agentic systems — verification pipelines, fallback mechanisms, human-in-the-loop supervision — is not an optional extra; it is what makes a deployment defensible.

Roadmap

- 1 Part 1: The Pharma & Life Sciences Discovery Landscape
- 2 Part 2: Robin – An End-to-End Discovery Loop
- 3 Part 3: The Agent's Scientific Toolbelt
- 4 Part 4: Teaching Agents to Reason — RL with Verifiable Rewards
- 5 Part 5: Privacy as a Deployment Constraint
- 6 Part 6: A Local, Private Agent in Practice
- 7 Part 7: myGenAssist — a Deep Agent in Production
- 8 Part 8: Conclusion**

Conclusion: discovery *and* deployment, both bounded

The discovery story

- multi-agent teams now produce **wet-lab-validated** hypotheses (Robin, Virtual Lab, Co-Scientist)
- the LLM orchestrates; **deterministic open tools** (Boltz, HF models) do the science
- credible wins are **narrow**; the human + wet lab is the truth boundary

The deployment story

- sensitive medical data demands **local / private** models and de-identification at the source
- a full agentic pipeline can run **with the network off**
- the same patterns ship in **real production** (myGenAssist), where the **replayable trajectory** is the unit of QA

The takeaway, sharpened for life sciences

The goal is not maximum autonomy. It is **appropriate autonomy under explicit constraints** — and in medicine, those constraints are **the wet lab** (truth), **the network boundary** (privacy), and **trajectory-level evaluation** (trust).

Discussion prompts

- For your discovery question, what is the **smallest autonomy level** that still adds value — and where must a human gate the funnel?
- Which steps genuinely need a **wet lab** to decide truth, and which can stay in-silico?
- For your clinical data, what must **never leave the building** — and could a local de-id gate make a cloud step safe?
- When is a small **local** model (extraction / de-id) the right tool, and when do you actually need a frontier reasoner?
- How would you **evaluate** an AI data analysis you can't fully reproduce (recall Robin's 7.5× vs. 1.75×)?
- Of the **open research challenges** (scalability, lifelong adaptation, explainability, secure communication, embodied/multimodal agents), which is the biggest blocker in *your* domain?

Take-home principles (life sciences)

- ❶ **The wet lab is the ground truth.** Agents generate and rank; experiments decide.
- ❷ **LLMs can't grade their own science.** Expert + experimental validation is non-negotiable.
- ❸ **Tools do the science; the LLM orchestrates.** Value lives in deterministic scientific tools.
- ❹ **Narrow + validated beats broad + in-silico.** Calibrate claims to what was confirmed.
- ❺ **Privacy is an architecture, not a promise.** De-identify at the source; keep PHI on-prem.
- ❻ **The trajectory, not the answer, is the unit of QA.** Replayable traces turn failures into permanent regression tests.
- ❼ **Appropriate autonomy, bounded by lab, network, and monitoring.** That is the whole game.

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